

Seiberg-Witten Invariants (A topologist's perspective)

X - closed, oriented 4-manifold

$$SW_X : \text{Spin}^c(X) \rightarrow \mathbb{Z}$$

Smooth invariants $X \cong_{C^\infty} X' \Rightarrow SW_X = SW_{X'}$

Defined by counting solutions to the
SW eqs:

$$D_A \chi = 0 \quad (\text{Dirac eqn})$$

$$F_A^2 = \sigma(\chi) \quad (\text{Monge eqn})$$

Definition

Defⁿ A spin^c -structure on X is a lift

$$\begin{array}{ccc} BU(1) & \rightarrow & B\text{Spin}^c(4) \\ & \nearrow & \downarrow \\ M & \xrightarrow{\tau} & BSO(4) \end{array}$$

where $\text{Spin}^c(4) = (\text{Spin}(4) \times U(1)) / \pm (I, 1)$

"complexified spin group"

(central extension of $SO(n)$ by $U(1)$)

$$\text{Spin}^c(X) := \{ \text{Spin}^c\text{-structures on } X \}$$

By general nonsense,

$$\begin{aligned} \text{Spin}^c(X) &\xleftrightarrow{!} [X, BU(1)] \cong [X, \mathbb{C}P^\infty] \\ &\cong [X, K(\mathbb{Z}, 2)] \\ &\cong H^2(X; \mathbb{Z}) \end{aligned}$$

geometrically, $\xi \in \text{Spin}^c(X)$ determines Not Canonical

- pair of rank 2 complex vector bundles

$$\xi \in [X, B\text{Spin}^c(4)] \quad \xi^+, \xi^- \rightarrow X \quad \text{"Spinor bundles"}$$

$$\cong \{ \text{Spin}^c(4) \text{ p.l./} X \}, \quad S = \xi^+ \oplus \xi^- := P_{\text{Spin}^c(4)} X \times_{\Delta} \mathbb{C}^4$$

- complex line bundle $L := \frac{\det(\xi)}{\det(\xi^+) \det(\xi^-)} \cong$ "fundamental representation"

- Clifford multiplication $\rho: TX \rightarrow \text{Hom}(\xi^+, \xi^-)$

Fact $\text{Spin}^c(X) \neq \emptyset$

Fix $\xi \in \text{Spin}^c(X)$ & a metric on X .

Defⁿ For A a $U(1)$ -connection on L ,

get a covariant derivative

$$\nabla^A : \Gamma(S^+) \rightarrow \Gamma(S^+ \otimes T^*X)$$

by coupling w/ Levi-Civita connection

The Dirac operator coupled w/ A is

$$D_A = \rho \circ \nabla^A : \Gamma(S^+) \rightarrow \Gamma(S^-)$$

Defⁿ $(A, \chi) \in \left\{ \begin{array}{c} U(1)\text{-connections} \\ \text{on } L \end{array} \right\} \times \Gamma(S^+)$

The SW eq^{ns} are
$$\begin{array}{c} \parallel \\ A_\chi \end{array}$$

$$D_A \chi = 0$$

$$F_A^\pm = (\chi \otimes \chi^*)_{\uparrow} \quad \text{True free (self-dual) part}$$

$$\wedge^{2,+} \cong \otimes^2(S^+)$$

where $F_A^\pm$ is the self-dual part of the curvature of A , $F_A \in \Omega^2(X; i\mathbb{R})$.

SW eq^{ns} invariant under action of the gauge group $\mathcal{G} = \text{Map}(X, S^1)$.

$$u \cdot (A, \kappa) = (A - u^{-1} du, u \cdot \kappa)$$

The configuration space is $B_{\mathcal{G}} = \mathcal{A} / \mathcal{G}$.

The SW moduli space is

$$\mathcal{M}_X(\xi, g) := \{ \text{solutions} \} / \mathcal{G} \subset B_{\mathcal{G}}$$

Fact Can perturb by $\eta \in \Omega^+(X; i\mathbb{R})$
 $F_A^+ = \sigma(\kappa) + \eta$

So that $\mathcal{M}_X(\xi, g, \eta)$ is a smooth, compact, orientable manifold. ^(*)

$$(*) \quad b_2^+(X) > 0.$$

Apply Atiyah-Singer index theorem to get

Thm The expected dimension Euler char. Signature

$$d := d(\mathcal{M}(\xi)) = \frac{c_1(L)^2 - (2\chi(X) + 3\sigma(X))}{4}$$

A "homology orientation" induces an orientation of $M(\Sigma, g, \tau)$.

$\leadsto$ fundamental class $[M(\Sigma, g, \tau)] \in H_2(B_\Sigma^+; \mathbb{Z})$

Defⁿ The SW invts of X are given by

$$SW(\Sigma) := \left\langle \int_{\cap}^{\cap} M^{d/2}, [M(\Sigma, g, \tau)] \right\rangle$$

$$H^2(B_\Sigma^+; \mathbb{Z}) \cong H^2(\mathbb{CP}^\infty; \mathbb{Z})$$

(d even $\Leftrightarrow b_2^+(X)$ odd, else SW defined to be 0)

Main Theorems

Thm (Vanishing) Suppose $\pi_1(X) = 1$, $b_2^+ > 1$ odd

$SW_X \equiv 0$ if

- 1) $X = X_1 \# X_2$, $b_2^+(X_i) > 0$
- 2) X admits a metric of positive scalar curvature
- 3) X has an embedded sphere w/
 $[\Sigma]^2 \geq 0$, $0 \neq [\Sigma]$

Thm (Non-vanishing)

- 1) If X a s.c. complex surface,
 $b_2^+(X) > 1$, $SW_X(\pm c_1(X)) \neq 0$
- 2) (Taubes) (X, ω) symplectic, $b_2^+(X) > 1$,
 $SW_X(\pm c_1(X, \omega)) = \pm 1$
- ↗ define w.r.t. choice of
tame a.c.s.

Thm (Adjunction) $\Sigma \subset X$ embedded, oriented
connected surface w/ $[\Sigma]^2 \geq 0$. Then
 $\forall \xi \in \text{Spin}^c(X)$ s.t. $SW_X(\xi) \neq 0$,

$$2g(\Sigma) - 2 \geq [\Sigma]^2 + \langle \xi, [\Sigma] \rangle$$

Thm (Gromov Conjecture)

(Kronheimer-Mrowka, 1994)

Satisfies

$\Sigma \subset \mathbb{C}P^2$ smoothly embedded

$\Sigma = [C] \in H_2(\mathbb{C}P^2)$ a sm. alg. curve degree d ,

$$g(\Sigma) \geq \frac{(d-1)(d-2)}{2}$$

i.e. smooth algebraic curves are genus minimizing representatives.

Follows from adjunction.

Thm $\exists X, X'$ s.t. $X \cong_{C^0} X'$ but

$SW_X \neq SW_{X'}$, hence $X \not\cong_{C^\infty} X'$.

Example Computation

$$\Leftrightarrow [T] \in \text{Spin}^c(X)$$



Defⁿ Let $T^2 \subset X$ a torus w/ $[T]^2 = 0$

& $\kappa: S^1 \hookrightarrow S^3$ a knot. Then

Knot Surgery on T is as follows:

$$X_n = (X \setminus \nu(T)) \cup (S^1 \times (S^3 \setminus \nu(\kappa)))$$

SII
 $T^2 \times D^2$

$$S^1 \times S^1 \times \partial D^2 \hookrightarrow S^1 \times \partial D^2 \times S^1$$

where ∂D^2 identified w/ a longitude of κ .

(Morgan-Mrowka-Szabo '97)

Thm (Fintushtel - Stern '97) $b_2^+(X) > 1$, $\pi_1(X) = 1$
 $[T] \neq 0$ but $[T]^2 = 0$. Then $X \stackrel{\sim}{=}_{\mathbb{C}} X'$
and $SW_{X_k} = SW_X \cdot \Delta_k(t)$

This yields a plethora of exotica already.

E.g.

$E(1) := \mathbb{CP}^2 \# 9 \overline{\mathbb{CP}^2}$ admits an elliptic fibration
over P^1 w/ 6 singular fibers.
"Dolgachev surface"

Fact

For $\Sigma = [F] \in H_2(E(1), \mathbb{Z})$,

$$SW_{E(1), [F]} = 1$$

(vanishing thm for positive
Scalar Curvature +
Wall crossing formula)

Now, Surgery on $F \subset E(1)$ by a $T(p, q)$
torus knot yields a Dolgachev surface $E(1)_{p,q}$,
& if e.g., $p=2$, $q=3$,

$$SW_{E(1)_{2,3}} = 1 \cdot \Delta_{\mathcal{B}}(t) = t - 1 + t^{-1}$$

$$\begin{array}{l} \text{hence } E(1) \cong_{C^0} E(1)_{2,3} \\ \text{but } E(1) \not\cong_{C^\infty} E(1)_{2,3} \end{array} \left[\frac{(t^{p^4}-1)(t-1)}{(t^p-1)(t^t-1)} \right]$$

This was the first known exotic 4-mfld,
originally shown by Donaldson. ('87)

Weitzenböck formula:

$$D_A^2 \chi = \nabla_A^* \nabla_A \chi + \frac{S}{4} \chi + \frac{1}{2} F_A^+ \cdot \chi$$

For SW solutions,

$$0 = \nabla_A^* \nabla_A \chi + \frac{S}{4} \chi + \frac{1}{2} \sigma(\chi) \cdot \chi$$

$$\begin{aligned} \text{Then } 0 &= \int_X |\nabla_A \chi|^2 + \frac{S}{4} |\chi|^2 + \frac{1}{2} \langle \sigma(\chi) \cdot \chi, \chi \rangle \\ &= \int_X |\nabla_A \chi|^2 + \frac{S}{4} |\chi|^2 + C |\chi|^4 \end{aligned}$$

Since $S > 0$, integrand is positive, hence $\chi \equiv 0$.

i.e. reducible, so $\mathcal{M} = \emptyset$ for generic
metric & perturbation.

Acts like a mass term, ^{in Dirac equation}
forcing spinors